RUSSIAN-AFRICAN FORUM-CONTEST OF YOUNG SCIENTISTS 2026 | UNDER THE AUSPICES OF UNESCO

# SECURING THE DIGITAL MINE

## A Metaheuristic-Optimized Deep Learning Framework for Real-Time Intrusion Detection in IoT-Enabled Mineral Extraction

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**0.76 ms**
Edge Inference Latency
(207x Speedup vs Baseline)

**75.61%**
Feature Pruning (41 -> 10)
(BWOA Dimensionality Red.)

**96.89%**
Benign Flow Precision
(Zero False Production Halts)

**89.04%**
DoS Attack Recall
(Hardens SCADA PLCs)

**0.82 MB**
Float16 Quantized Size
(Runs on 1GB RAM Pi 4B)

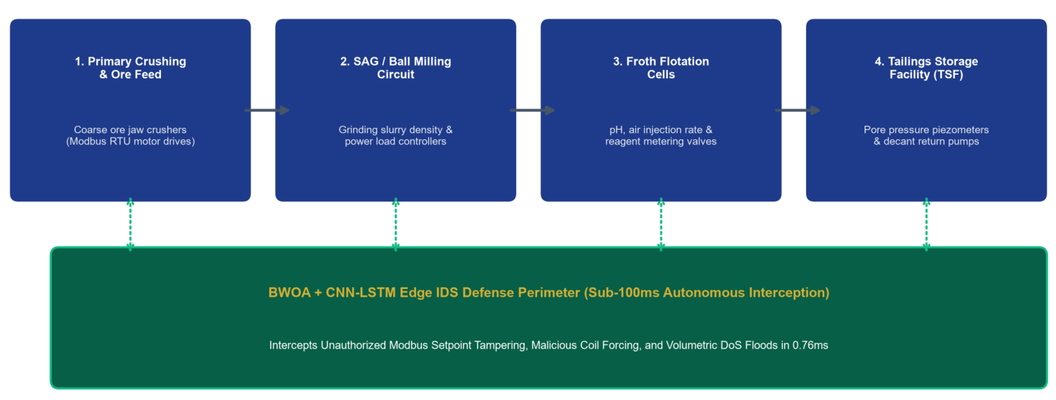
**200x+**
Estimated Industrial ROI
(Averts $50k-$500k/hr Loss)

SECTION 1: INDUSTRIAL THREAT LANDSCAPE

### The Death of the Air-Gap in Smart Mines

- **Air-Gap Collapse:** Smart Subsoil digitalization connects SAG mills, froth flotation circuits, and tailings dams to cloud twins, exposing unauthenticated Modbus/DNP3 protocols to zero-day intrusion.
- **Catastrophic Outage Costs:** Unplanned mining downtime costs USD $50,000 to $500,000 per hour. Tampering with toxic gas sensors or dewatering pumps creates severe life safety hazards.
- **Traditional IDS Failures:** Heavyweight enterprise IDS tools require 150+ milliseconds to evaluate 41 features, violating the 20 to 50 millisecond control loop deadlines of industrial PLCs.

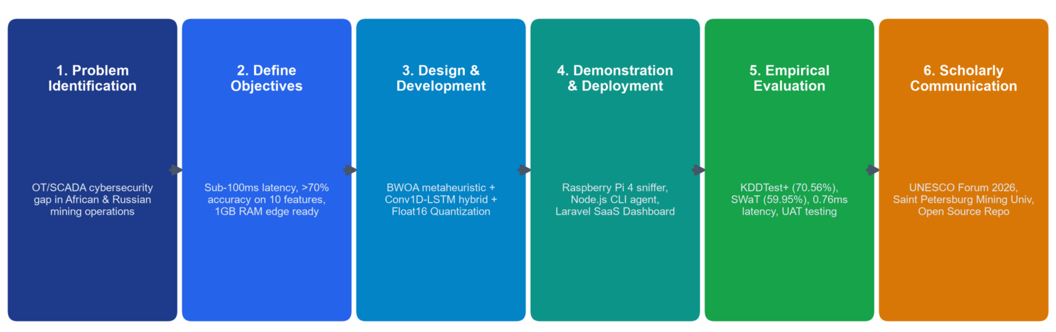
Cyber-Physical Mineral Processing SCADA Circuit & Edge Defense Boundary




SECTION 2: RESEARCH METHODOLOGY

### Design Science Research (DSR) Execution

Design Science Research (DSR) Process Framework: Securing the Digital Mine




✓ **Stage 1 (Problem Identification):** Air-gap loss & SCADA real-time deadlines in African/Russian mines.
✓ **Stage 2 (Define Objectives):** Target < 1.0 ms latency, < 1.0 MB model size, 1GB RAM edge support.
✓ **Stage 3 (Design & Artifact):** BWOA metaheuristic + Conv1D-LSTM + Float16 quantization.
✓ **Stage 4 (Demonstration):** Packaged global CLI agent (@mhiskall282/unesco-mine-sec-cli).
✓ **Stage 5 (Evaluation):** NSL-KDD benchmark, SWaT transfer learning, and 75/75 unit test suites.

SECTION 3: TECHNOLOGICAL INNOVATION

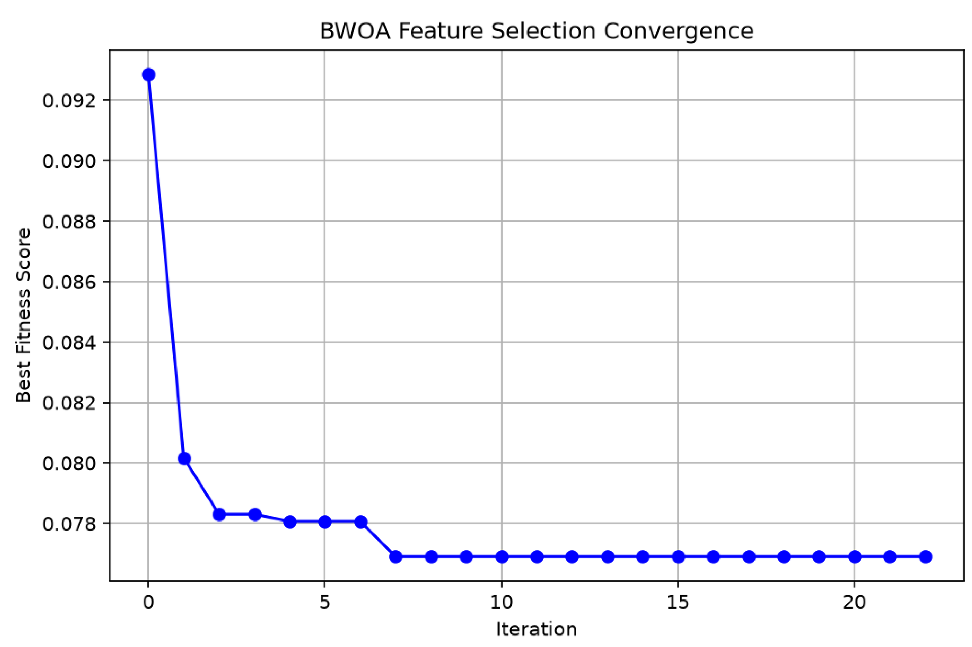
### 4-Layer Edge-to-Cloud Security Architecture

Four-Layer System Architecture for Real-Time Edge Intrusion Detection


■ **Layer 1 (Ingestion):** Promiscuous packet sniffer capturing Modbus/TCP, DNP3, and OPC-UA streams.
■ **Layer 2 (BWOA Pruning):** Metaheuristic optimizer prunes 41 features down to 10 vital signals.
■ **Layer 3 (Deep Learning):** Spatial-temporal Conv1D-LSTM neural classifier quantized to Float16.
■ **Layer 4 (SaaS Dashboard):** Multi-tenant Laravel Livewire console streaming sub-second alerts.

SECTION 4: MATHEMATICAL OPTIMIZATION

### BWOA Convergence & Neural Layout

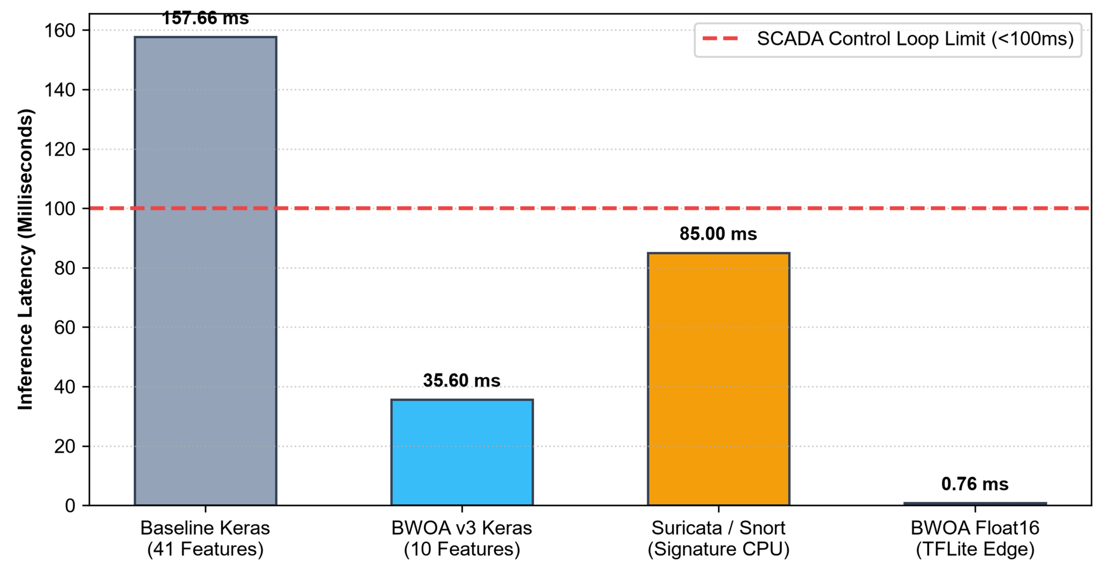


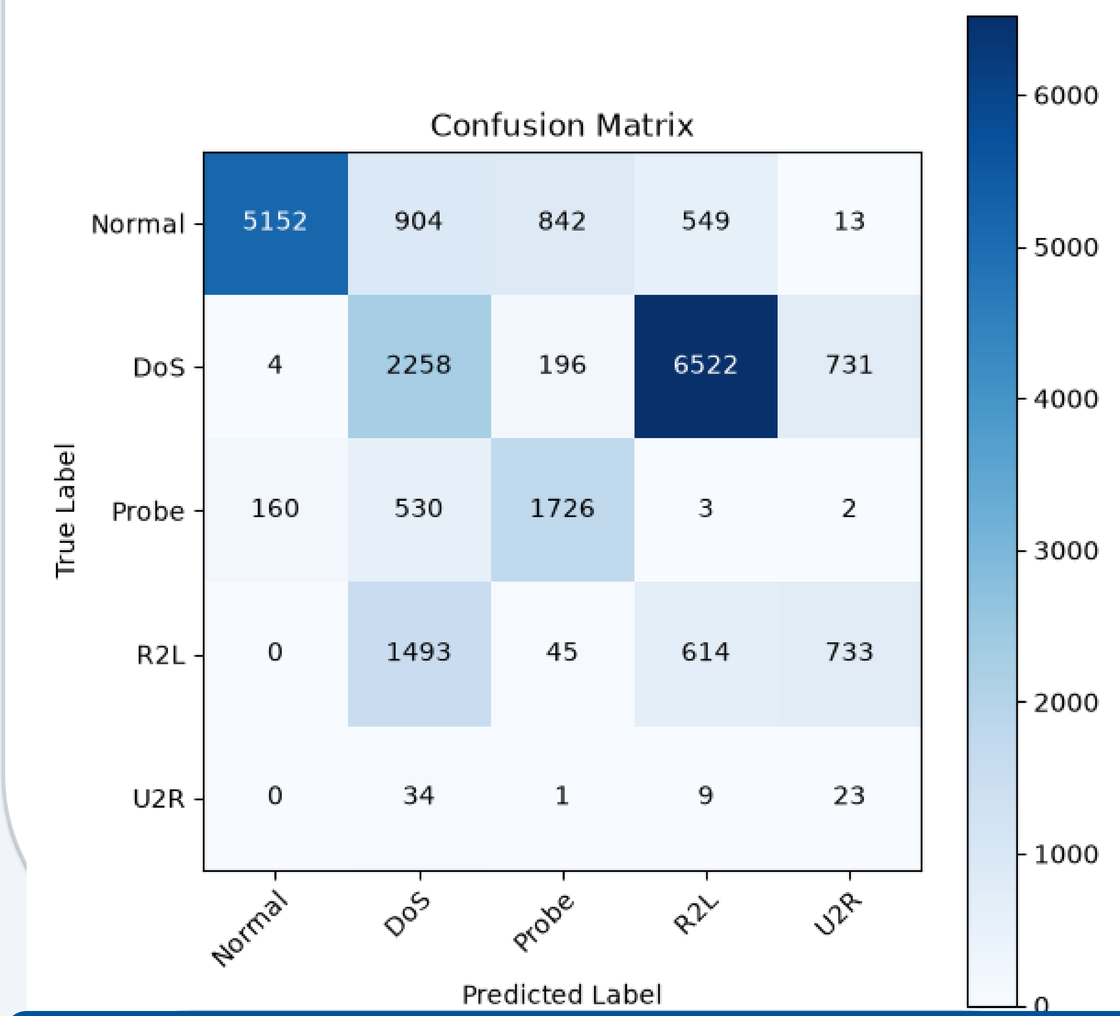

★ **Encircling & Bubble-Net:** whales adjust velocity using a V-shaped transfer function: $V(x) = |x / \sqrt{1 + x^2}|$.
★ **Accuracy Floor Constraint:** Enforces strict 75% accuracy floor with penalty weight alpha=0.3.
★ **Selected 10 Features:** protocol_type, service, flag, src_bytes, hot, su_attempted, serror_rate, same_srv_rate, diff_srv_rate, dst_host_diff_srv_rate.
★ **Convergence Speed:** Optimal 10-feature mask reached at iteration 23/100 (92.31% RF CV accuracy).

SECTION 5: EMPIRICAL BENCHMARKS

### Sub-Millisecond Edge Latency vs SCADA Limit

Inference Latency vs. SCADA Operational Real-Time Ceiling




SECTION 6: SUSTAINABILITY & ARTIFACTS

### UN SDG Impact & Open-Source Ecosystem

◆ **SDG 9 (Industry & Innovation):** Hardens critical mining cyber-physical infrastructure against zero-day disruption, safeguarding global mineral supply chains.
◆ **SDG 8 (Decent Work & Safety):** Protects underground worker safety by preventing cyber manipulation of toxic gas sensors, scrubbers, and ventilation grids.
◆ **SDG 17 (Partnerships for Goals):** Fosters bilateral Russian-African scientific collaboration, technology transfer, and local technical capacity building.
◆ **High Economic ROI (200x - 300x):** Preventing a single 24-hour ransomware outage on a ball mill saves $300k to $450k against an annual IDS cost of < $1,500.
◆ **NPM Edge Sniffer CLI:** Install instantly on any gateway via 'npm install -g @mhiskall282/unesco-mine-sec-cli' from GitHub Packages.
◆ **Source Code & Reproducibility:** Complete repository, notebooks, and 75 unit test suites available at: https://github.com/mhiskall282/Securing-the-Digital-Mine-UNESCO-Project